

Social Problem Spaces & Projects

PROJECT PRINCIPLE:

The projects are identifying a specific opportunity where technology could create meaningful leverage, then develop a reusable product core and demonstrate it as a working proof of concept (PoC) in one social context.

SOCIAL PROBLEM

Understand the people, context and constraints

USE CASE

Identify a specific situation in which a technical capability could help.

PRODUCT CORE PoC

Build a reusable module demonstrating the capability in one context – e.g., matching, retrieval, prediction, speech, optimization, or decision support.

FULL PRODUCTISATION & REUSE

Analyze what is needed to convert your PoC to a field ready product. Explore and identify other contexts and NGOs that could find use of developed technical capability.

What your PoC is—and is not

- The PoC should be the product core and be able to demonstrate a meaningful technical capability, not a complete production product.
- Open and synthetic data should be used to build and demonstrate your PoC.
- An NGO or social organisation provides problem context, workflow knowledge and feedback—not necessarily the dataset.
- A successful PoC should be evaluated quantitatively where possible, with clear limits on what the evaluation does and does not establish.
- You should state what additional data, validation, human processes and infrastructure would be required before real-world deployment.

Important: A model working on an open dataset does not automatically establish that it will work in the field. Understanding that gap is part of the learning objective.

Connecting People to the Right Support, Services & Opportunities

THE SOCIAL PROBLEM

People often know that they need help but struggle to identify what support exists, whether they are eligible, which option is most relevant, or what they should do next. Information is often fragmented across programmes, organisations and sources.

PROJECTS

01A

Finding relevant support

How might we help a person identify and prioritise the services, programmes or opportunities that are most relevant to their circumstances and needs?

01B

Finding the next best step

How might we help a person navigate from a stated need or goal to an appropriate next action, while identifying information that is still missing?

TECHNICAL CAPABILITY AREAS

- Recommendation & ranking
- Eligibility reasoning
- Semantic search / retrieval
- Knowledge graphs
- Conversational information gathering

SOCIAL & NGO CONTEXT

Career guidance, scholarships, skilling, social services, women's support, health services.

Direct anchors: Nirmaan, Barabari Collective and Umeed for Women.

PoC DATA SOURCES

01A

Starting sources:

[myScheme](#)
[India Government Schemes](#)

Synthetic/self-generated:

Generate synthetic citizen profiles and derive eligibility/ground truth from scheme rules.

Using this source:

Official scheme pages should remain the source of truth.

01B

Starting sources:

[myScheme](#)
[OGD Platform India](#)

Synthetic/self-generated:

Generate synthetic needs, incomplete profiles and expected next actions.

Using this source:

Treat recommendations as a benchmark, not proof of real-world suitability.

Reaching People Excluded by Language, Literacy or Digital Interfaces

THE SOCIAL PROBLEM

Information may technically be available but remain inaccessible because people cannot comfortably read, type, navigate or interact through the available interface. The challenge is to make access possible through modes people can actually use.

PROJECTS

02A

Natural-language access

How might we enable people to access essential information and services through the language and mode of interaction they are most comfortable with?

02B

Making information understandable

How might we transform complex information into explanations that are understandable and actionable for people with different language, literacy and communication needs?

TECHNICAL CAPABILITY AREAS

- Speech recognition / synthesis
- Translation & transliteration
- Conversational AI
- RAG
- Information simplification

SOCIAL & NGO CONTEXT

Healthcare information, education, government services, career guidance and blood-donation coordination.

Direct anchors: Blood Warriors

PoC DATA SOURCES

02A

Starting sources:

[Common Voice](#)
[AI4Bharat IndicVoices](#)
[Samanantar](#)

Synthetic/self-generated:

Generate domain queries, transliteration variants and conversations; do not rely on synthetic speech alone for ASR evaluation.

Using this source:

Public speech data may not capture local accents, noise, code-mixing or domain vocabulary.

02B

Starting sources:

[OGD Platform India](#)
[myScheme](#)
[WHO open access](#)

Synthetic/self-generated:

Generate complex/simple pairs, questions and controlled comprehension cases.

Using this source:

Check factual preservation; readability alone is not sufficient.

Helping People Learn More Effectively

THE SOCIAL PROBLEM

Teachers and programmes cannot always identify who is struggling, what they are struggling with, and what support would help next. This is especially challenging when learning happens across online, asynchronous and in-person settings.

PROJECTS

03A

Identifying who needs learning support

How might we identify learners or groups that need additional attention using learning activity and assessment information already available?

03B

Identifying what a learner needs next

How might we infer a learner's current understanding and recommend what they should learn, practise or revisit next?

TECHNICAL CAPABILITY AREAS

- Knowledge tracing
- Learner modelling
- Learning-gap detection
- Personalisation
- Intervention prioritisation

SOCIAL & NGO CONTEXT

College skilling, school education and vocational learning.

Direct anchors: Barabari Collective and Nirmaan.

PoC DATA SOURCES

03A

Starting sources:

[OULAD](#)
[KDD Cup 2010](#)
[UCI Student Dropout](#)

Synthetic/self-generated:

Generate longitudinal learner histories with controlled performance and engagement patterns.

Using this source:

Datasets come from particular education settings; do not claim universal learner-risk prediction.

03B

Starting sources:

[EdNet](#)
[OULAD](#)
[KDD Cup 2010](#)

Synthetic/self-generated:

Generate learners with latent mastery states and known concept dependencies.

Using this source:

EdNet is from an AI tutoring service in Korea; transfer to Indian learners must be tested separately.

Moving from Education & Skills to Livelihoods

THE SOCIAL PROBLEM

People may have education, skills or aspirations but struggle to understand which skills matter, which opportunities fit them, and what pathway connects their current capabilities to meaningful work.

PROJECTS

04A

Mapping skills to pathways

How might we help an individual understand the skills they have, the skills they are missing, and possible pathways towards meaningful livelihood opportunities?

04B

Matching people to opportunities

How might we match individuals to jobs, training, mentorship or other livelihood opportunities while accounting for both capabilities and circumstances?

TECHNICAL CAPABILITY AREAS

- Skill extraction
- Competency mapping
- Semantic matching
- Knowledge graphs
- Pathway recommendation

SOCIAL & NGO CONTEXT

Career guidance, skilling, employment and disability-inclusive livelihoods.

Direct anchors: Nirmaan and Barabari Collective.

PoC DATA SOURCES

04A

Starting sources:

[O*NET](#)
[ESCO](#)
[ILOSTAT](#)

Synthetic/self-generated:

Generate candidate profiles with current skills, missing skills and aspirations.

Using this source:

O*NET/ESCO are not India-specific; add Indian context separately.

04B

Starting sources:

[O*NET](#)
[ESCO](#)
[PLFS Microdata](#)

Synthetic/self-generated:

Generate candidate and opportunity records with skills, location, education and constraints.

Using this source:

A synthetic opportunity set is preferable to dependence on scraped commercial job portals.

Making Sensitive Conversations More Constructive & Inclusive

THE SOCIAL PROBLEM

Some conversations are difficult to facilitate because participants may be uncomfortable, disengaged, dominate the conversation, misunderstand one another, or introduce exclusionary or harmful perspectives.

PROJECTS

05A

Supporting facilitators

How might we help facilitators recognize what is happening in a sensitive group conversation and decide when and how intervention may be useful?

05B

Making conversations more reflective

How might we help participants and facilitators identify patterns, perspectives and missed opportunities in a conversation that are difficult to notice in real time?

TECHNICAL CAPABILITY AREAS

- Conversation analysis
- Moderation
- Bias / toxicity detection
- Perspective extraction
- Dialogue assistance

SOCIAL & NGO CONTEXT

Youth development, inclusion, community dialogue and education.

Direct anchors: Umeed for Women

PoC DATA SOURCES

05A

Starting sources:

[AMI Meeting Corpus](#)
[Civil Comments](#)
[Social Bias Frames](#)

Synthetic/self-generated:

Generate controlled dialogues with dominance, interruption, disagreement, harmful statements and interventions.

Using this source:

These are technical proxies, not representations of real gender/inclusion workshops.

05B

Starting sources:

[MELD](#)
[AMI Meeting Corpus](#)
[Social Bias Frames](#)

Synthetic/self-generated:

Generate conversations with known perspectives, missed viewpoints and outcomes.

Using this source:

MELD is derived from scripted television dialogue; do not claim field validity.

Helping People Adopt & Sustain Beneficial Behaviours

THE SOCIAL PROBLEM

Many social outcomes depend not simply on whether people know what they should do, but whether they adopt and sustain behaviours over time. Support may need to adapt to different behavioral patterns.

PROJECTS

06A

Understanding behavioral patterns

How might we identify patterns in people's behaviour that help explain when they adopt, disengage from, or sustain a beneficial activity?

06B

Adapting support to individuals

How might we determine what kind of intervention or support may be appropriate for different behavioural patterns, rather than providing the same intervention to everyone?

TECHNICAL CAPABILITY AREAS

- Behaviour sequence modelling
- Clustering
- Personalisation
- Churn / dropout prediction
- Intervention recommendation

SOCIAL & NGO CONTEXT

Physical activity, programme participation, education and health-related behaviors.

Direct anchor: Hyderabad Runners.

PoC DATA SOURCES

06A

Starting sources:

[UCI Human Activity Recognition](#)
[PAMAP2](#)
[WISDM](#)

Synthetic/self-generated:

Longitudinal sequences with engaged, declining and recovering states.

Using this source:

Activity-recognition data is not programme-adherence data; associations are not causal evidence.

06B

Starting sources:

[StudentLife](#)
[UCI Human Activity Recognition](#)
[PAMAP2](#)

Synthetic/self-generated:

Generate user trajectories and intervention responses.

Using this source:

Real intervention outcomes are required to establish that an intervention changes behavior.

Helping Frontline Workers Prioritise & Act

THE SOCIAL PROBLEM

Frontline workers often face more cases, information and decisions than they can effectively process within available time. Technology can help them notice what matters, prioritise attention and navigate information without replacing human judgement.

PROJECTS

07A

Identifying cases that need attention

How might we help frontline workers identify which cases require attention first when they have limited time and incomplete information?

07B

Supporting frontline decisions

How might we help frontline workers gather relevant information, understand a case and identify appropriate next actions without replacing human judgement?

TECHNICAL CAPABILITY AREAS

- Case prioritisation
- Information extraction
- Decision support
- Evidence retrieval
- Agentic workflows

SOCIAL & NGO CONTEXT

Health, education, livelihoods and community programmes.

Direct anchors: Barabari Collective and Blood Warriors.

PoC DATA SOURCES

07A

Starting sources:

- [OGD Platform India](#)
- [WHO open access](#)

Synthetic/self-generated:

Generate cases with urgency indicators, missing information and defined priority labels.

Using this source:

Synthetic labels encode assumptions; expose those assumptions and test robustness to missing/noisy information.

07B

Starting sources:

- [OGD Platform India](#)
- [WHO open-access myScheme](#)

Synthetic/self-generated:

Create synthetic cases with gold-standard evidence and expected actions.

Using this source:

Guidelines cannot cover every real case; systems should expose uncertainty and escalate outside their evidence base.

Connecting Scarce Resources to People Who Need Them

THE SOCIAL PROBLEM

Many social programmes operate with scarce resources: donors, blood, mentors, training slots, scholarships, staff, equipment or service capacity. The challenge is deciding who should receive what, when and where.

PROJECTS

08A

Matching scarce resources

How might we match limited resources to people who need them in a way that is timely, fair and effective?

08B

Anticipating resource needs

How might we anticipate where and when demand for a scarce social resource is likely to arise so that organisations can prepare before shortages occur?

TECHNICAL CAPABILITY AREAS

- Matching
- Optimisation
- Fair allocation
- Demand forecasting
- Simulation / scheduling

SOCIAL & NGO CONTEXT

Blood donation, scholarships, mentors, healthcare capacity and training.

Direct anchors: Blood Warriors

PoC DATA SOURCES

08A

Starting sources:

[UCI Blood Transfusion Service Center](#)
[OGD Platform India](#)

Synthetic/self-generated:

Generate recipients, inventory, availability, compatibility, distance and urgency.

Using this source:

The UCI dataset is donor-focused, not a real blood-demand/inventory dataset; simulate the allocation problem.

08B

Starting sources:

[OGD Platform India](#)
[ILOSTAT](#)

Synthetic/self-generated:

Generate inventories, demand seasonality, spikes, lead times, expiry and capacity.

Using this source:

Forecast quality depends strongly on the assumed demand-generating process.

Making Fragmented Information Useful for Decisions

THE SOCIAL PROBLEM

People and organisations often have important information spread across documents, reports, guidelines, records, messages and databases. The challenge is turning fragmented information into something useful for a specific decision while retaining evidence.

PROJECTS

09A
Bringing fragmented information together

How might we combine information from multiple sources to create a coherent picture of a situation without losing the evidence behind it?

09B
Turning information into actionable insight

How might we help a person identify the information that matters most for a particular decision while making the reasoning and evidence traceable?

TECHNICAL CAPABILITY AREAS

- RAG
- Information extraction
- Entity resolution
- Knowledge graphs
- Evidence synthesis / provenance

SOCIAL & NGO CONTEXT

Direct anchors: Hyderabad Runners. Also Applicable to all our other guest NGOs.

PoC DATA SOURCES

09A
Starting sources:
[OGD Platform India](#)
[WHO open-access](#)
[OpenAlex](#)

Synthetic/self-generated:
Create synthetic multi-document cases with facts distributed across sources.

Using this source: Public documents may contain inconsistent names, dates and definitions; conflict resolution should be evaluated.

09B
Starting sources:
[OGD Platform India](#)
[WHO open-access](#)
[OpenAlex](#)

Synthetic/self-generated:
Create synthetic decision cases with gold-standard evidence and expected conclusions.

Using this source: Evaluate retrieval, evidence coverage, citation correctness and decision accuracy separately.

Enabling People with Disabilities to Participate More Independently

THE SOCIAL PROBLEM

Many systems assume that everyone can interact through the same combination of sight, hearing, speech, typing and conventional interfaces. People with disabilities may need alternative or adaptive modes of interaction.

PROJECTS

10A

Alternative ways of interacting

How might we enable people with different abilities to interact with information and services through modes that work for them?

10B

Adapting interaction to the individual

How might digital systems adapt their interaction based on how a person is able to communicate, perceive and respond?

TECHNICAL CAPABILITY AREAS

- Sign-language / gesture recognition
- Speech
- Computer vision
- Multimodal interaction
- Adaptive interfaces

SOCIAL & NGO CONTEXT

Education, employment, public services and digital accessibility.

Direct Anchor: Ashray Akruthi.

PoC DATA SOURCES

10A

Starting sources:

[INCLUDE Indian Sign Language](#)
[INCLUDE code and models](#)

Synthetic/self-generated:

Use open data for training and create a small self-recorded test set.

Using this source:

INCLUDE covers isolated signs and specific signers; do not present it as general ISL translation.

10B

Starting sources:

[INCLUDE Indian Sign Language](#)
[Common Voice](#)

Synthetic/self-generated:

Generate synthetic capability profiles, then validate modality selection with a small self-collected test set.

Using this source:

Real users are needed to establish whether the chosen interaction actually improves accessibility.